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Validity

This document is a DRAFT documentation for Sensor XT.

Sensor XT is not officially released but only launched as an MVP (Minimal Viable Product) for pilot installations of selected customers.

Because of this, the documentation is not released on the homepage but only made available on request. Further information about Sensor XT can also be found at Sensor XT MVP SSI MLT 20 C BT (internal user access to Lightweb needed).

If you find mistakes, have questions or feedback, please contact your local TRIDONIC support.

TRIDONIC GmbH & Co KG is constantly striving to develop all its products. This means that there may be changes in form, equipment and technology.

Claims cannot therefore be made on the basis of information, diagrams or descriptions in these instructions.

Copyright

This documentation may not be changed, expanded, copied or passed to third parties without the prior written agreement of TRIDONIC GmbH & Co KG.

We are always open to comments, corrections and requests. Please send them to info@tridonic.com

Imprint

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About the device



Sensor XT is a sensor for smart city streetlight applications.

With the included image-based sensor it is possible to monitor streets, places and parking lots and collect information about the occupancy and the number of passing objects, both people and vehicles.

The three main applications are

- parking lot management
- traffic analysis and
- people and vehicle presence detection.

The gained information can be used by owners, planners or other organizations. To protect people's privacy, pictures are only processed inside the sensor and not transferred via DALI interface. So, no critical information is accessible from the outside.

Sensor XT is made to be used with luminaires compatible with the Zhaga18 standard. This ensures an easy installation of the device over a wide range of existing luminaires from various manufacturers.

For an easy setup of Sensor XT in the field, there is a Bluetooth® interface included together with a dedicated Android mobile app. This allows the user to go through the whole process of setting the device up and configuring the functions in a plain and user-friendly way.

Different sensor modes

Sensor XT can be used in two different sensor modes called “Application controller” and “Monitoring detection”. These sensor modes differ as follows:

- “Application controller”
Standalone operation with integrated application controller
 - Only local control of the directly connected DALI drivers is possible
 - All connected drivers react identical (DALI Broadcast)
 - No additional DALI-2 sensor to be connected
 - Available functionality in this operation mode:
 - Switch the light on or off based on presence (either people or cars) -or-
 - Dim the light to preset levels depending on traffic density (number of cars per time)

Notice:

The two described functionalities in this operation mode can only be used separately. A combination of them is not possible.

- “Monitoring detection”
Device acts as an input device connected to a separate application controller supporting DALI-2 control device Instance Type 3 (presence) and/or Instance Type 6 (general purpose) for full functionality.
Information that can be obtained from sensor for processing in application controller:
 - Parking lot management
 - Traffic analysis
 - People and vehicle presence detection

Notice:

All this information can be obtained in parallel for each zone/crossing line.

Further information about how to set up and commission the sensor modes can be found in the following chapters:

- Selecting the desired sensor mode
- Commissioning “Application controller” mode -and-
- Commissioning “Monitoring detection” mode.

First steps in the app

Sensor XT comes with a dedicated Bluetooth® app. The following first steps in the app are the same for both sensor modes “Application controller” and “Monitoring detecting”.

Installing the app

Notice:

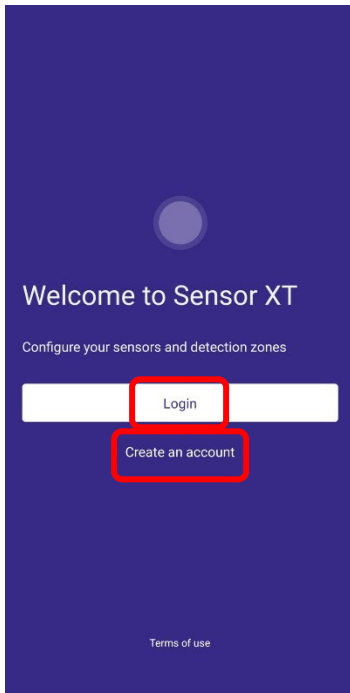
During the pilot phase of Sensor XT, the app is only available for the Android mobile operating system and can only be obtained directly from Tridonic.

The app is not available for other operating systems like Apple iOS and the available Android version of the app cannot be downloaded directly from the Google app store.

- Contact your local Tridonic support to receive information about how to download the app.
- Install the app.

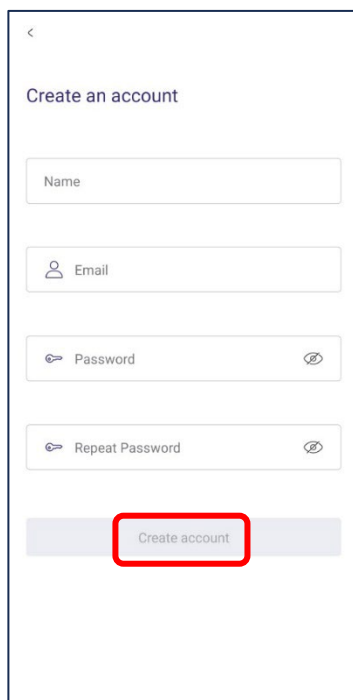
Creating an account

→ Opening the app for the first time, the following screen is shown:



- Click “Login” if you have already created an account
-or-
click “Create an account” to create an account.

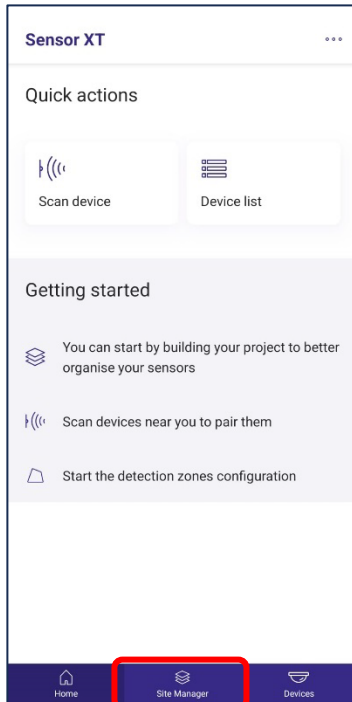
→ After a click on “Create an account”, the following screen is shown:

A screenshot of the 'Create an account' form. The background is white. At the top, there's a back arrow and the title 'Create an account'. Below the title, there are four input fields: 'Name', 'Email', 'Password', and 'Repeat Password'. Each field has a small icon on the right side. At the bottom, there is a grey button labeled 'Create account', which is highlighted with a red rectangular box.

- Enter the required data (name, email, password).
- Click “Create account”.

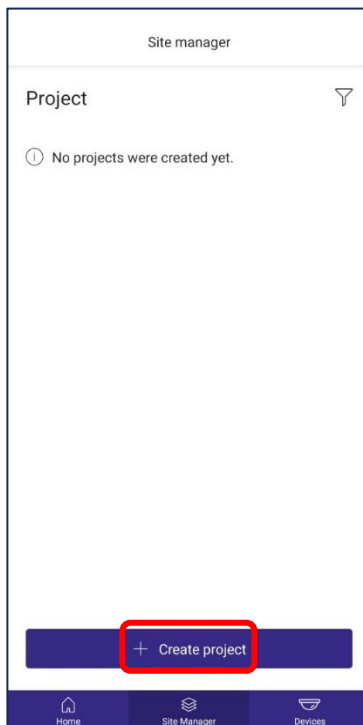
Creating a project

→ After creating an account and logging in, the following screen is shown:



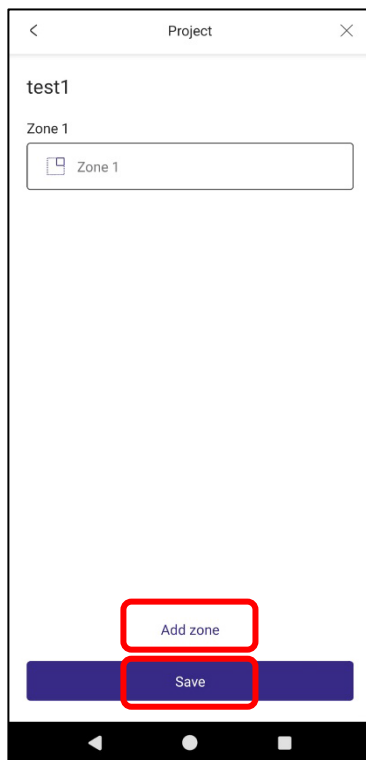
- Click “Site Manager”.

→ The following “Site manager” screen is shown:



- Click “Create project”.

→ After creating a project with e.g. name “test1”, the following “Project” screen is shown:



- Click “Add zone” to add the zones where the sensors are placed.
- Click “Save” to save project and zones.

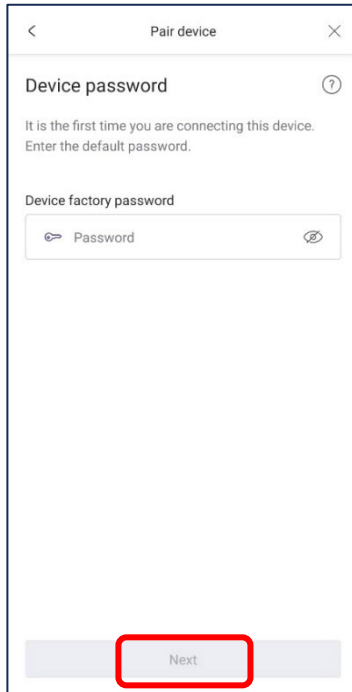
Notice:

Zones offer the possibility to divide a project into sub-areas. These sub-areas can be assigned to one or more sensors.

Adding a device to the project

- Click “Scan device” on home screen.
If required, give the needed rights to the app for Bluetooth® technology and positioning.

→ The following “Pair device” screen is shown:



- To add a device, follow the instructions in the GUI.
- Click “Next” in-between steps.

Notice:

There are two types of passwords in the app:

- Account password: for the app as a whole
- Device password: for individual sensors

The default “Device factory password” is “password”.
Enter this default password when connecting for the first time
and create your own personal password afterwards.

→ The following “Project” screen is shown:



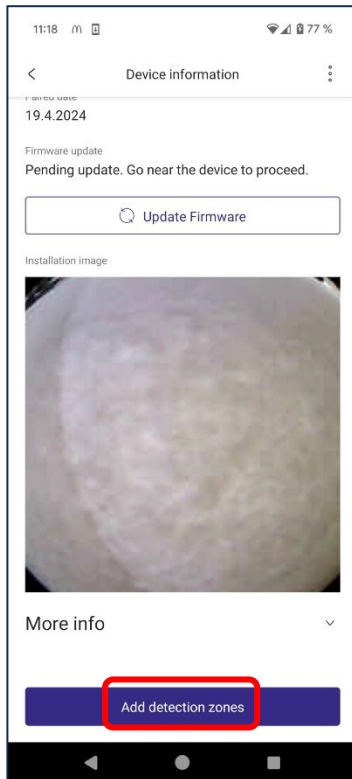
- Click “Add zone” to add a zone.
- Follow the instructions on the screen to define zones.
- Click “Save” to upload the defined zones to the mobile device.

Notice:

Zones offer the possibility to divide a project into sub-areas.
These sub-areas can be assigned to one or more sensors.

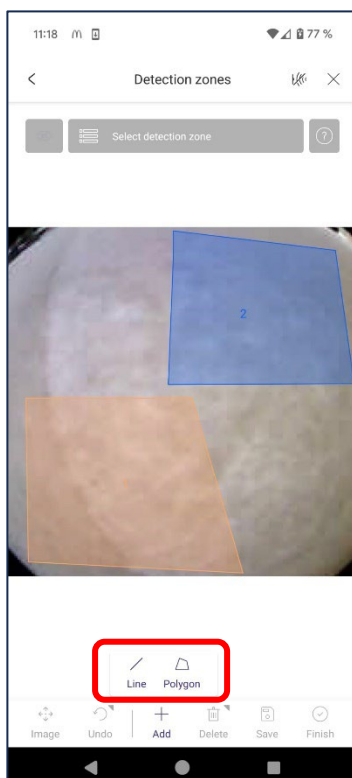
Adding a detection zone

- Go to “Devices” menu at bottom of screen.
- Select the desired device.
→ The following “Device information” screen is shown:

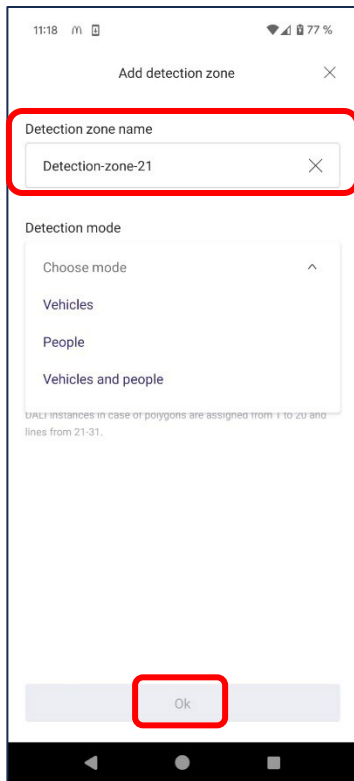


- Click “Add detection zones”.

→ The following “Detection zones” screen is shown:

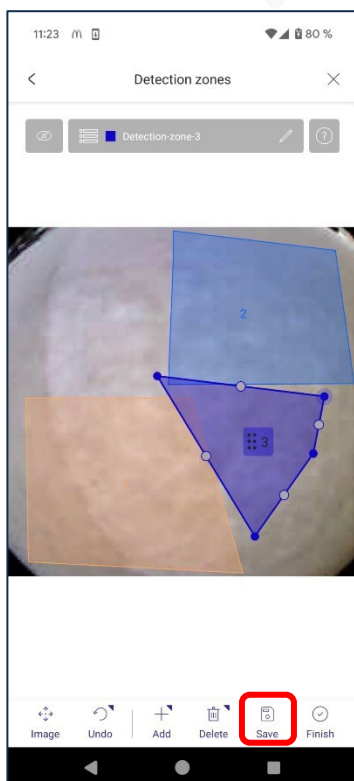


- Select the desired shape for the detection zone:
 - “Line” is for traffic analysis.
 - “Polygon” is for presence detection and parking lot management.



- Define a "Detection zone name".
- Select the desired "Detection mode" from the drop-down menu.
- Click "Ok".

→ The following "Detection zones" screen is shown:



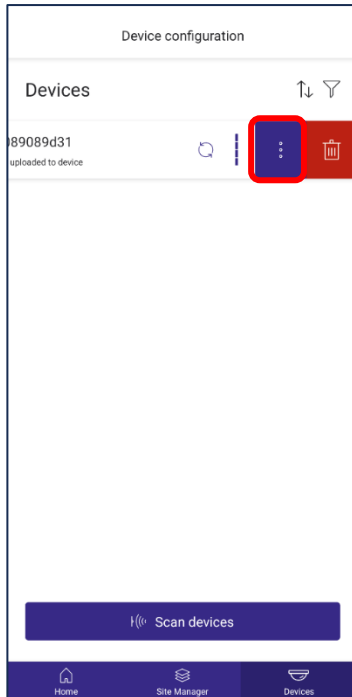
- Draw a detection zone.
- Click "Save" to save.
- Click "Upload" to upload to the sensor.

Selecting the desired sensor mode

The following chapter describes how to set the “Sensor mode” to either “Application controller” or “Monitoring detection”.

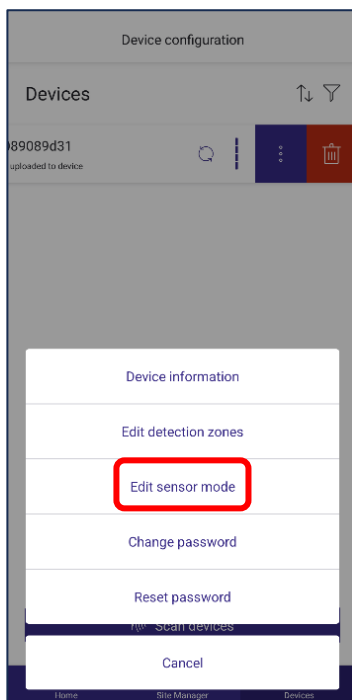
- Go to the device configuration by selecting “Devices”.
- Go to “Devices” menu at bottom of screen.

→ The following screen is shown:



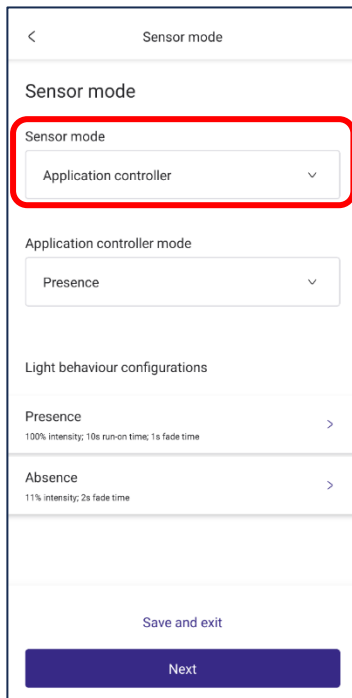
- Select the desired device.
- Swipe to the left.
→ The 3-dots menu is shown.
- Click on the 3-dots menu to open it in a separate window.

→ The following screen is shown:



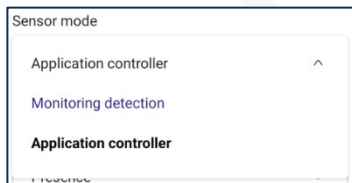
- Click “Edit sensor mode”.

→ The following screen is shown:



- Click on the drop-down menu “Sensor mode” to open it.

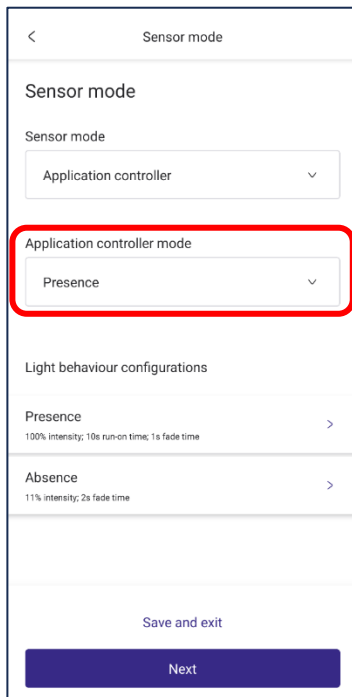
→ The drop-down menu for “Sensor mode” is shown:



- Select “Application controller” or “Monitoring detection”.
- Click “Save and exit” if done
- or-
- move to next step for the commissioning of the selected “Sensor mode”.

Commissioning of “Application controller” mode

→ After having set the “sensor mode” to “Application controller”, the following screen is shown:



Sensor mode

Sensor mode

Application controller

Application controller mode

Presence

Light behaviour configurations

Presence

100% intensity; 10s run-on time; 1s fade time

Absence

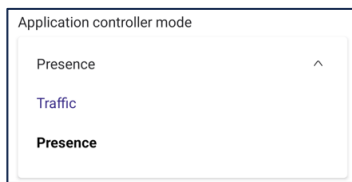
11% intensity; 2s fade time

Save and exit

Next

- Click on the drop-down menu “Application controller mode” to open it.

→ The following sub menu for “Application controller mode” is shown:



Application controller mode

Presence

Traffic

Presence

You can select between “Traffic” and “Presence”.

If you select “Presence”, the following screen is shown:

Daylight configuration

On threshold

1 lux 1000 lux 1000

Off threshold

150% 2000% 1890 %

Off threshold 1890% = 18900 lux

Fade time

1 s

Save and exit

Upload configurations

- Adjust the available settings for “On threshold”, “Off threshold”, “Fade time”.
- Click “Save and exit” to save your changes
-or-
click “Upload configurations” to load configurations to the sensor.

The parameters on this page define different thresholds as explained in the text. More information about “Presence detection” can be found below in chapter Presence detection parameters.

Notice:

Be aware that the sensor detects presence the same way in all detection zones defined for this sensor.

- Presence state is entered if at least one detection area is reporting “presence”.
- Absence state is entered once all detection zones are reporting “absence” simultaneously.

If you select “Traffic”, the following screen is shown:

Traffic density

Low to medium Medium to high

10 60

Set the threshold values of vehicles per 10 minutes.

Fade time

Lower to higher

0 s

Higher to lower

0 s

Set the threshold values from lower to higher values and from higher to lower values.

Lower light intensity

0% 100% 5 %

Medium light intensity

0% 100% 25 %

Higher light intensity

0% 100% 83 %

- Adjust the available settings for “Traffic density”, “Fade time”, the “Light intensity” settings, “Daylight configuration”.
- Click “Next” in-between steps.
- Click “Save and exit” to save your changes locally on your mobile device
-or-
click “Upload configurations” to upload configurations to the sensor.

The parameters on this page define different thresholds as explained in the text. More information about “Traffic density” can be found below in chapter Traffic density parameters.

Daylight configuration

On threshold

1 lux 1000 lux 1000

Off threshold

150% 2000% 1890 %

Off threshold 1890% = 18900 lux

Fade time

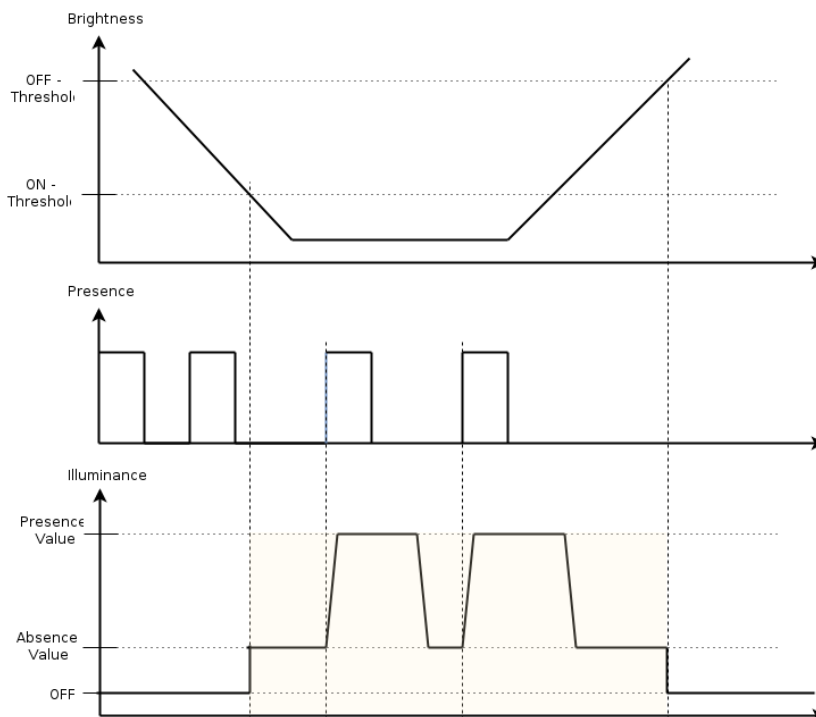
1 s

Save and exit

Upload configurations

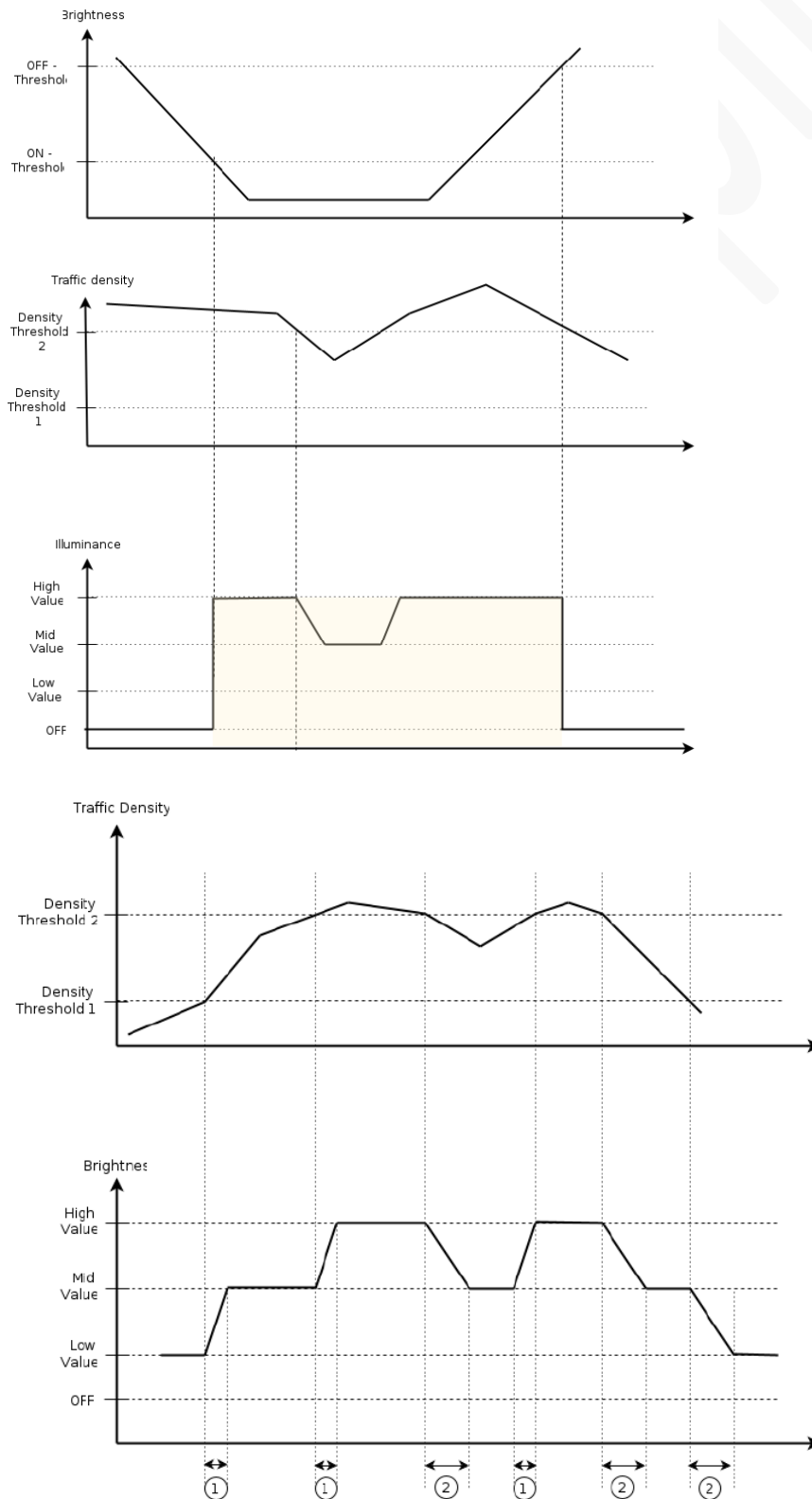
Presence detection parameters

The Enhanced Presence Detection AC enables the illuminance to be linked to presence or absence of people and it is based on corridorFUNCTION. Presence linking is depending on the ambient brightness and is only active if the value is below a certain threshold - otherwise the illuminance is 0% (OFF).



Traffic density parameters

The Traffic Monitoring enables the illuminance to be linked to the traffic density. Traffic density linking is depending on the ambient brightness and is only active if the value is below a certain threshold - otherwise the illuminance is 0% (OFF).



1. Fade time: time during which the intensity is changing from a lower to an higher value
2. Fade time: time during which the intensity is changing from a higher to an lower value

Commissioning “Monitoring detection” mode

Make sure that the “Sensor mode” is set to “monitoring detection” as described in chapter Selecting the desired sensor mode.

Notice:

For the commissioning in the monitoring detection mode there are at least two commissioning processes that need to be executed before Sensor XT is working.

- Step1: commissioning of Sensor XT (detection zones) -and-
- Step 2: commissioning of Sensor XT into a CMS System.

For step 2, depending on the used system, the commissioning for “Monitoring detection” can differ and depends on the kind of system and its documentation.

The following text only provides general important information needed for the commissioning and the use of Sensor XT.

DALI instance types for people and vehicle presence detection

The following chapter describes the supported instance types and numbers that Sensor XT communicates to DALI.

DALI instances used by Sensor XT

When used in “monitoring detection”, Sensor XT acts as a DALI input device with up to 32 DALI instances. These are structured as follows:

Instance Number	Instance Type	Description	Reference applicable standard
0	4	Light Sensor	IEC 62386-304 Ed.1
1	3	Presence Detector Zone 1	IEC 62386-303 Ed.1
2	3	Presence Detector Zone 2	IEC 62386-303 Ed.1
...	...	...	...
20	3	Presence Detector Zone 20	IEC 62386-303 Ed.1
21	6	General Purpose Sensor Zone 1	IEC 62386-306 Ed.1
22	6	General Purpose Sensor Zone 2	IEC 62386-306 Ed.1
...	...	...	...
31	6	General Purpose Sensor Zone 11	IEC 62386-306 Ed.1

Although there are 32 instances available, the amount of instances used in Sensor XT is determined during commissioning with the Sensor XT Bluetooth® app. During commissioning the amount, type and position of detection zones is set, corresponding instances are enabled, "unused" instances remain / are set to disabled. The process of enabling/disabling instances is managed in Sensor XT internally.

The above applies to all instances, except for Instance #0 (LightSensor), which is enabled per default.

Instance type description

Instance Type 4 - Light Sensor

Instance Type 4 refers to light sensor functionality as described in IEC 62386-304 Ed.1

Sensor XT is capable of detecting different illuminance levels and report these via DALI.

Instance Type 3 - Presence detector

Instance type 3 means that the instance has the functionality specified in IEC 62386-303 Ed.1 "Input devices - Occupancy sensor".

A presence detection zone is a polygon which can be configured individually per instance. Zones can be added via the Sensor XT Bluetooth® app described above. The Sensor XT is capable of detecting different object types. (e.g. people, cars). The objects that trigger a presence event can be configured separately per instance.

Depending on the configuration, presence events are triggered by one or multiple object types entering/leaving a presence detection zone.

The following values are transmitted:

- 0x00: Vacant and no movement
- 0xAA: occupied and no movement.

Notice:

Depending on environmental conditions (lighting conditions, shadows, contrast, ...) and increased processing, there might be a short delay (a few seconds) before proper presence state is reflected on DALI.

Instance Type 6 - General Purpose "Traffic density"

Instance type 6 means that the instance has the functionality specified in IEC 62386-306 Ed.1, "Input devices - General purpose sensor".

The aim is to provide an information about how many objects on average are passing by the detection zone in a period of time ("traffic density")

The detection zone is a crossing line. Crossing lines can be defined in the Sensor XT Bluetooth® app the same way as zones. Objects crossing that line - regardless of their direction - are counted. The Input Value generated by this, shall make the information how many objects on average have been passing by in a period of time, available on the DALI-bus.

The Sensor XT is capable of detecting 2 different object types: People and cars. The object type that triggers the object crossing can be configured separately per instance. Object type vehicles comprises 'light' (cars) and 'heavy' (trucks, buses) vehicles. When a crossing line is configured to detect both cars and people, the sum of the two counters is then used to calculate the traffic density for the corresponding instance.

Definition of Measurement unit and quantity name according to IEC 62386-306 Ed.1 Annex A shall be as follows:

- "unitofMeasurement" is "0" (manufacturer defined). It shall be called: "objects per 10 min"
- "quantityName" for input value is "0" (manufacturer defined). It shall be called: "Traffic Density".

Instance Type 6 – Variables

The following variables apply to all Instance Type 6 instances of Sensor XT unless stated otherwise.

Variable	Default value	Range of validity	Comments
'resolution'	n.a.	9	Size of resolution due to max. 9 bit event message size.
'magnitude'	127	'magnitude' = 'magnitudePhMax' = 'magnitudePhMin' = 127	
'inputSignalled'	0	0	
'inputValue'	0x00	[0x0000....0x01FE], 0xFFFF	
'measuredValue'	0x00	[0x00.... 0x01FE]	(no offset --> K=0)
'unitofmeasurement'	0	0	Definition Sensor XT: "objects per 10 min"
'unitofmeasurementextended'	0	0	
'quantityName'	0	0	Definition Sensor XT: "traffic density"
'quantityNameextended'	0	0	

Additional remarks on instance type traffic density:

The time period for the calculation of objects/period is 10 minutes.

This implies that reliable values of "traffic density" can be obtained earliest 10min after detection for the selected instance has started. (E.g. power-up)

The report timer (specified in IEC 62386-306) is used to send the "trafficDensity" in a DALI event from the Sensor XT to the DALI application controller. The accuracy of the report timer has a tolerance of at least +/-1 %.

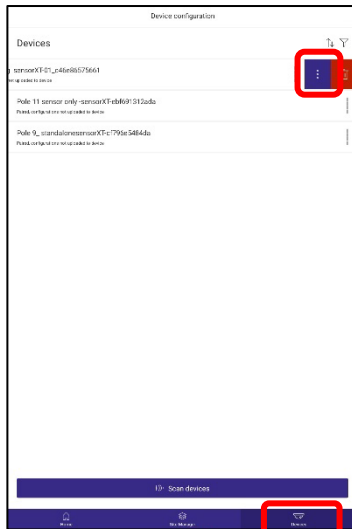
The maximum "value" that fits in a DALI event for a 306 Instance type is 511dec (9 Bit) .

There is a possibility to trigger separate events in case a set threshold value gets crossed.

This can be achieved by using so called "Alarm events" as described in IEC 62386-306 Ed.1. The threshold value can be chosen individually, trigger can be surpassing or falling below threshold. Up to 4 different Alarms are possible.

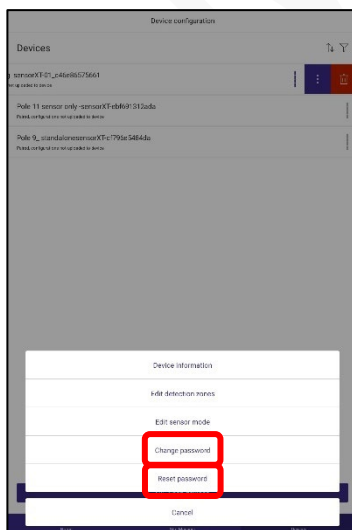
Resetting options

Change or reset password



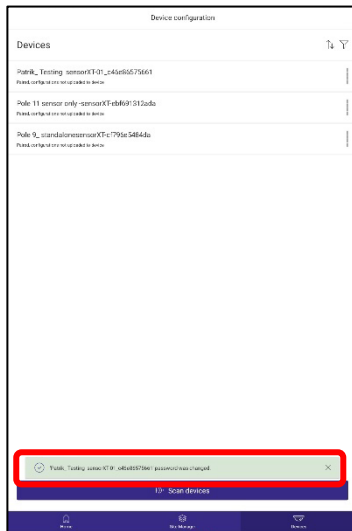
- Click “Devices.”
→ The “Device configuration” screen is shown.
- Go to desired device and swipe left.
→ The three-dot menu is shown.
- Click on three-dot menu.

→ The parameters of the three-dot menu is shown:

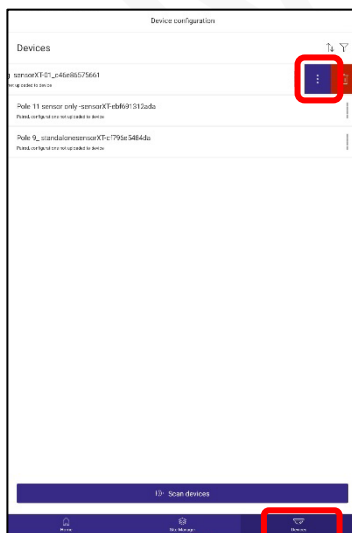


- Click “Change password”
-or-
click “Reset password”.
- Follow the steps described in the GUI.
- Click “Next” in-between steps.

→ After changing or resetting the password the “Device configuration” screen with a confirmation message is shown.

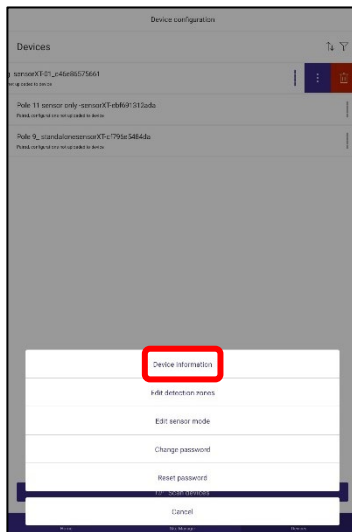


Factory reset



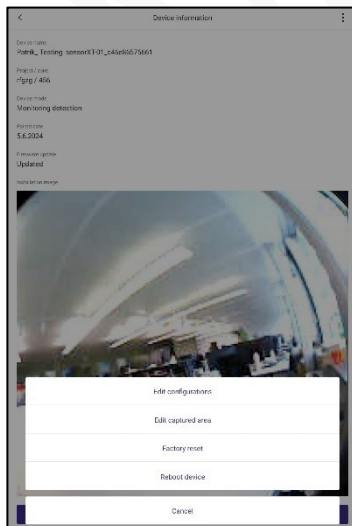
- Click “Devices.”
→ The “Device configuration” screen is shown.
- Go to desired device and swipe left.
→ The three-dot menu is shown.
- Click on three-dot menu.

→ The parameters of the three-dot menu is shown:



- Click “Device information”.

→ The next “Device information” screen is shown:



- Click the three-dot menu at the top.
→ The parameters of the three-dot menu are shown.
- Click on “Factory reset”.
- Follow the steps described in the GUI.
- Click “Next” in-between steps.

DALI standard deviations

Although equipped with a ZHAGA Book 18 socket, the Sensor XT is neither Di4 nor DALI-2 certified as of today because it is a pilot project. In addition to that there are some exemptions to "standard" DALI behaviour. In principle the Sensor XT supports commands and functionality according to the following DALI standards:

- IEC 62386-103 Ed.1 + AMD1
- IEC 62386-303 Ed.1
- IEC 62386-304 Ed.1
- IEC 62386-306 Ed.1
- DiiA Specification Part 351 V1.0

Deviations IEC 62386-103

Parameter/Command/ Functionality affected	Description of deviation
ENABLE APPLICATION CONTROLLER (OP-CODE 0x16) DISABLE APPLICATION CONTROLLER (OP-Code 0x17)	• Command is ignored, status of Variable "applicationActive" is not changed. • Status of "applicationActive" variable can be queried, but reported value is always "FALSE" and therefore not valid.
ENABLE INSTANCE (OP-Code 0x62) DISABLE INSTANCE (OP-CODE 0x63)	• Although supported, it is strongly advised to NOT make use of these commands. It will most likely lead to inconsistencies with Sensor X internal configuration and may very likely lead to unexpected behaviour. • per default all instances are disabled. • Status of all "instanceActive" variables is valid, it can be queried as usual. Notice: Enabling/Disabling of instances is done by configuring the detection zones via the Sensor XT mobile application.
SET OPERATING MODE (DTR0) (OP-Code 0x18)	• "operatingMode" is per default 128 • Range of validity is [0 or 128]. • Change of operating mode affects format of input value of Light sensor events (acc. IEC 62386-304 Ed.1) • Status of "operatingMode" variable is valid, it can be queried as usual.

Deviations DiiA Specification 351

- Sensor XT is classified as "Type B" device.
- Electrical requirements for AUX Supply are met. (externally powered, AUX max. 1W)

Parameter/ Command/ Functionality affected	Description of deviation
Arbitration algorithm	Arbitration algorithm is not implemented. Enable/Disabling of internal AppCon only done via Sensor XT Bluetooth® App during commissioning.
Memory Bank 201 implementation	Addr. 0x07 reporting value 0x00 although it is a "Type B" controller. Value is read only, cannot be changed via DALI.

Installation and environment

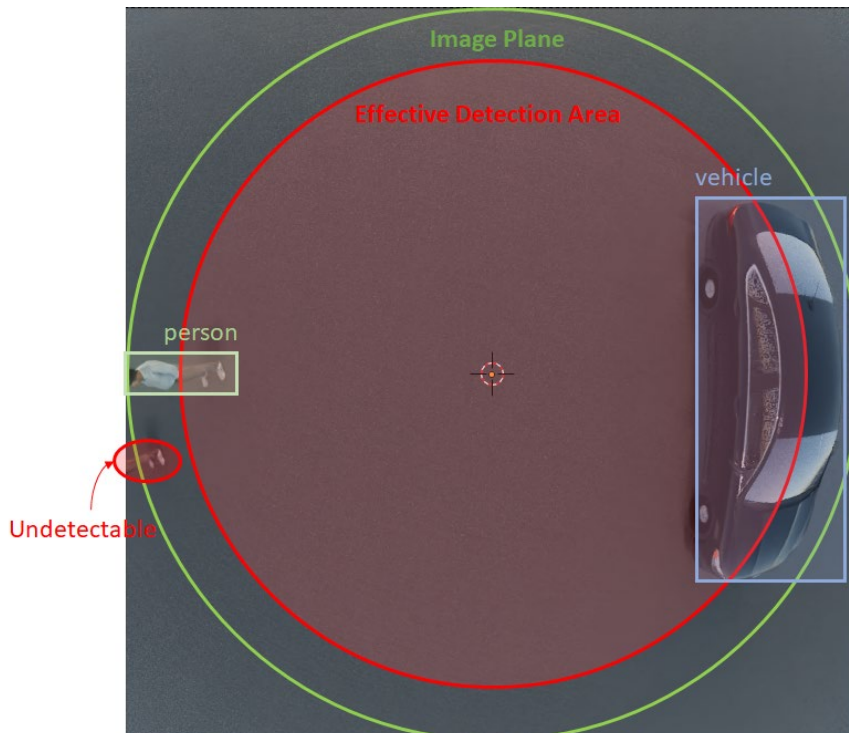
General Installation/ Mounting

- Maximum mounting tilt +/-10 to perpendicular line
→ Please check with data sheet.
- The effective detection area to be covered is directly dependent on the mounting height.
→ The higher the sensor is mounted the larger the effective detection area can be.
- Unplanned motion of the mounting structure (e.g. light pole swinging in wind) may have influence on detection accuracy.

General considerations on operating conditions

Detection Area

- Overall detection area is circular around an imaginary, (perpendicular) line going from the sensor centre to the ground.
- Tilting of the sensor will result in lower resolution of more remote areas of image.
→ This may lead to lower detection accuracy and/or delayed detection, especially for higher mounting heights.
- Compared to the image plane, the effective radius/area where object detection is possible is reduced.
 - Only objects which are visible in full can be detected.
 - (The major part of) detection zones/ crossing lines must be placed in the effective detection area.
 - Human average height assumed 1.8 m
 - Vehicle average height assumed 1.5 m
- In case of fast moving objects (traffic vehicles), it needs to be considered that vehicles can be detected in a wider range than persons.



Conclusions

- Overall detection accuracy is higher in the image center then in the edge area.
→ Preferably have detection areas/crossing lines close to the image center wherever possible.
- Recommendation is to have few, but larger detection areas.
→ Ensuring objects are fully covered.

Optical conditions

As Sensor XT is an image based sensor it requires some considerations with regard to influences on the optical system:

- It is imperative to ensure a minimum environmental light level during all operation. This light level is 3 lux. This implies that operation of Sensor XT is not possible in total darkness.
- An uninterrupted direct line of sight between sensor and detection area is necessary at all times.
- The performance of the detection algorithm may be impacted by changes of environmental optical conditions which may vary significantly over time. Examples for such conditions are natural and artificial lighting levels, uniformity of lighting levels, shadows, contrast (object vs. background, background only), object colours, ...)

Conclusions

As far as possible try to ensure the following:

- Proper lighting conditions during all operating times. E.g. street lighting in operation avoiding disturbing objects blocking line of sight to intended detection zones.
- Ensure clean, transparent lens.
- Consider temporarily discarding sensor info in case it is foreseeable that optical conditions will be bad.

Hardware installation commissioning

People presence commissioning

The system can accept commissioning within the radius presented in Figure 1.

For installation heights over 6 m an active radius of 10 m (round area of 314 m²) is set as the upper limit the user can commit any area.

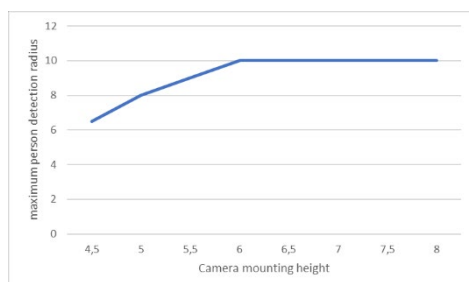


Figure 1: The radius within a crossing line can be placed for people presence.

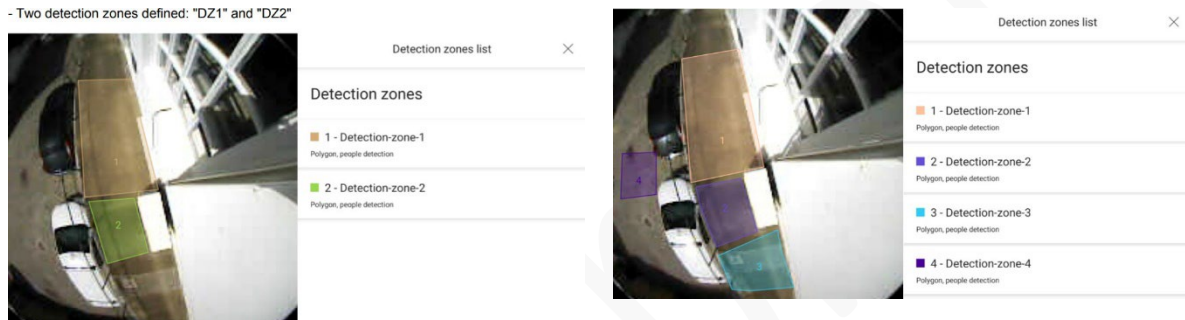


Figure 2: Initial tests 4 Detection Zones

4 Detection Zones (DZ) were commissioned, as shown in Figure 2: Initial tests 4 Detection Zones.

Suggestions

- A basic rule of thumb should be creating a commissioning area of 2.0 m x 2.0 m that could fit 9 people standing close to each other in 3x3 formation, as shown in Figure 3. It seems that DZ4 is smaller than this suggestion.

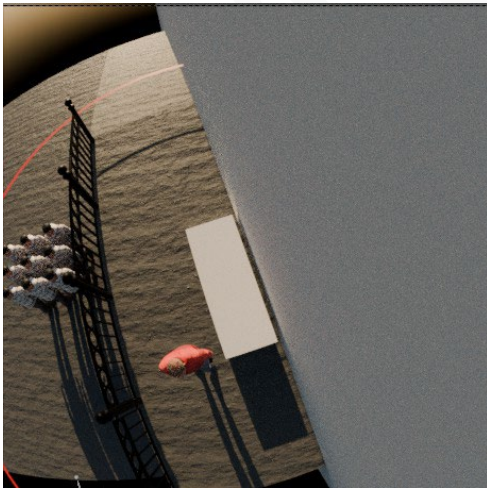


Figure 3: Commissioning suggestion

Furthermore, try not to commission area in the border line of the given image as in low installation heights situations (i.e. 4.5 m). This may not work properly.

Traffic counting commissioning

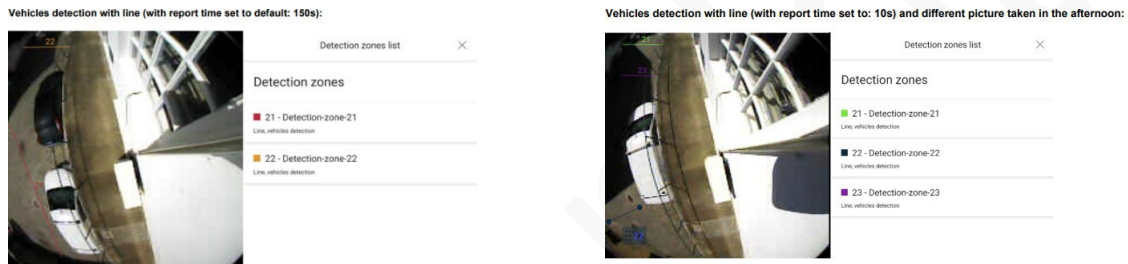


Figure 4: Initial Detection Zones

Considering a traffic flow line that moves onto the image plane, commissioning lines should cut this line vertically, as shown in Figure 5.

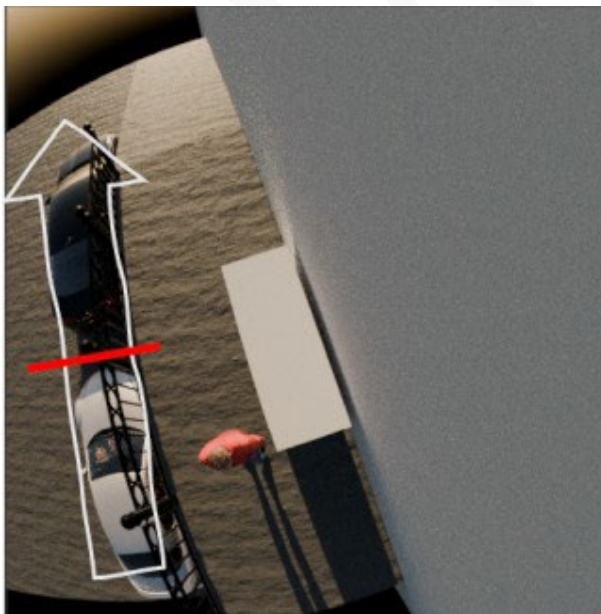


Figure 5: Commissioning line for traffic counting

Suggestions

- Try to draw the line in the middle of the traffic lane (flow) inspected by the camera module.
- The commissioning line should be vertical to the traffic flow.
- A rule of thumb is that a car should fit before and after the commissioning line in order for the system to work properly.

Parking inspection commissioning

Coverage assumption 1:

All computations assume average car dimensions of 4.8 m length, 2 m width and 1.5 m height.

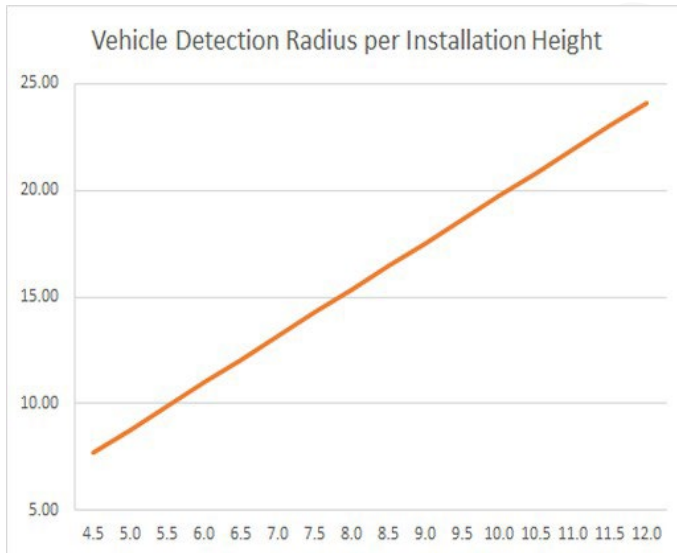
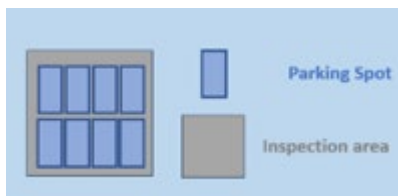


Figure 6: Vehicle detection radius per camera installation height in meters.

As shown in Figure 6 and assuming that the average vehicle (car) height is set to 1.5 m, the car detection radius varies per installation height.

Coverage assumption 2:

With the Parking Spot Dimensions set to 2.5 m × 5 m we calculate the maximum parking spots that fits (can be commissioned) in the vehicle detection area at a given camera mounting height.



Coverage assumption 3:

The upper detection limit is set at 15 m distance to camera mounting position.

Notice:

Adjacent parking spots introduce physical occlusions when occupied by vehicles. Those occlusions affected by camera mounting height, car height and parking spot distance to the camera position.

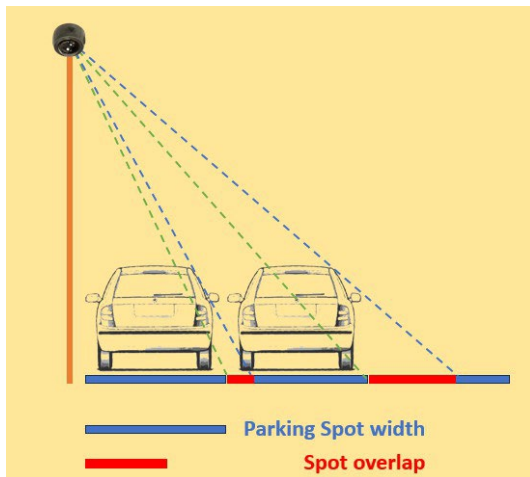


Figure 7: Overlap example

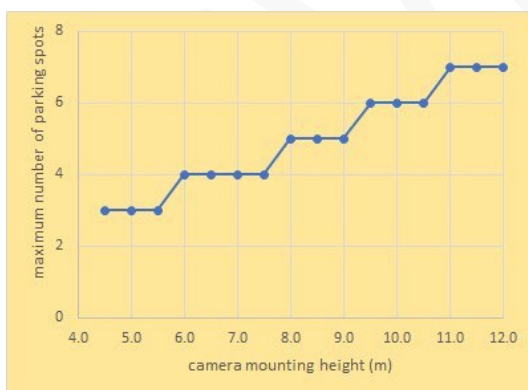


Figure 8: Number of parking spots, depending on mounting height

For an average car height of 1.5 m, the maximum number of adjacent parking spots per side, that can be inspected by the system, are shown in the given figure. Those numbers are independent of camera module and affected only by the physical overlap introduced by the parked car dimensions.

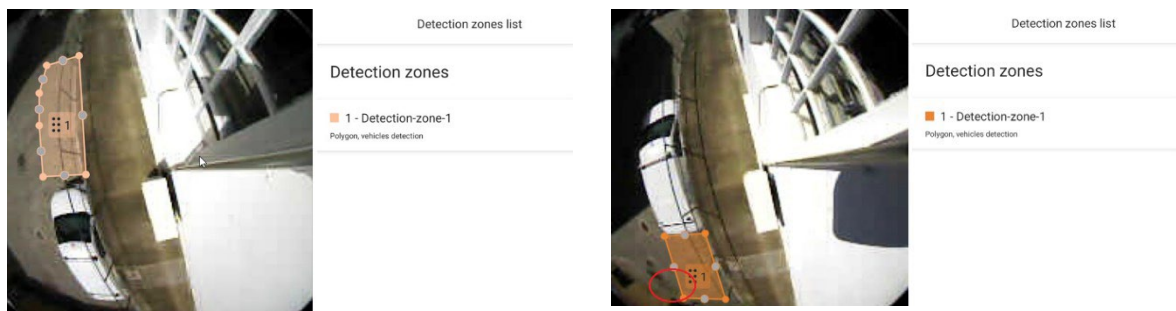
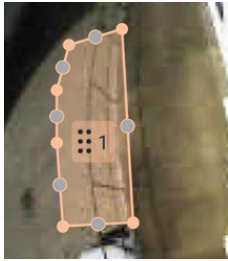


Figure 9: Initial parking lots commissioning.

Suggestions

- Try to create detection zones, where a car fits in, not smaller ones like the commissioned area presented as “bad practice” in Figure 10.

Figure 10: Good practice vs. bad practice on parking commissioning.



Good practice



Bad practice

Status LED

There is a single red LED built-in to indicate the status information to the user.

Implemented functionality:

- LED flashes 5x when power is applied to the device to indicate proper supply
- When the Dali-2 command “Identify device” is sent to the device, the LED flashes for 10 seconds.



Tipps and Tricks

Manual factory reset

Physical access reset mechanism

- If it is not possible to access the device using Bluetooth® communications to do the device reset or the user does not have yet paired with the device, the user must have physical access to the device to do the reset to factory;
- The reset mechanism is done using Power-On/Power-Off cycles of the sensor;
- Sensor device uses the same Factory reset mechanism used in Tridonic Matter devices

The following table shows the duration of the power cycles:

Stage	Duration	State
1	<5 s	ON
2	>30 s	OFF
3	5-15 s	ON
4	>30 s	OFF
5	<5 s	ON
6	>30 s	OFF
7	<5 s	ON
8	>30 s	OFF
9	permanent	ON

The following image shows the power cycles in a graphic style:



Reconnect Bluetooth

It may occur that when selecting “scan device” in the Sensor XT Bluetooth® app, that the sensor XT might not be found, although it is powered and within radio communication distance. (~20m)

Solution description:

- Try to close the app and restart it. (See <https://support.google.com/android/answer/9079646?hl=en#zippy=%2Cclose-apps>)
- You may need to reenter password when re-connecting to sensor.

Selecting the correct Sensor XT in case several devices are shown.

In case there are several sensor within reach and shown in the Sensor XT Bluetooth® app at the same time, it can be difficult to pick the correct one directly.

Ideally in such a case only the sensor to be commissioned is powered.

Additionally you can distinguish sensors by the signal strength shown by 4 vertical dots. See following figure.

10:02 Mo., 10. Juni 22%

Device configuration

Devices	
Patrik_Testing sensorXT-01_c46e86575661 Paired, configurations uploaded to device	
Pole 11 sensor only -sensorXT-ebf691312ada Paired, configurations uploaded to device	
Pole 9_ standalonesensorXT-cf796e5484da Paired, configurations not uploaded to device	

Note: Signal strength indication needs to be updated manually by rescanning devices.

Factory reset: Save 'Device UUID' and "DSP serial number"

In case the sensor needs to be reset to factory default, it is necessary to enter 'Device UUID' and "DSP serial number" to be able to execute a factory reset or a password reset.

Normally these values are entered automatically in case you have an established BT connection to the sensor.

In case you don't have an established connection to the device (e.g. using new tablet for reconfiguration) it may occur that you need to enter 'Device UUID' and "DSP serial number" manually. To ensure information is at hand it is recommended to copy this information during commissioning.

Solution description:

- After or during commissioning select the "device information" screen with the picture provided by sensor.
- Open the menu "more info" below the picture.
- Copy the 'Device UUID' and "DSP serial number" by using the copy symbol on the right and save this information to a text file or Email.
- Save this information safely on an external storage / send Email to ensure information is available at later stage.
- It is recommended to add additional information like chosen device name, site name/ configuration date/ commissioning engineer incl. contact details etc. to the same file.